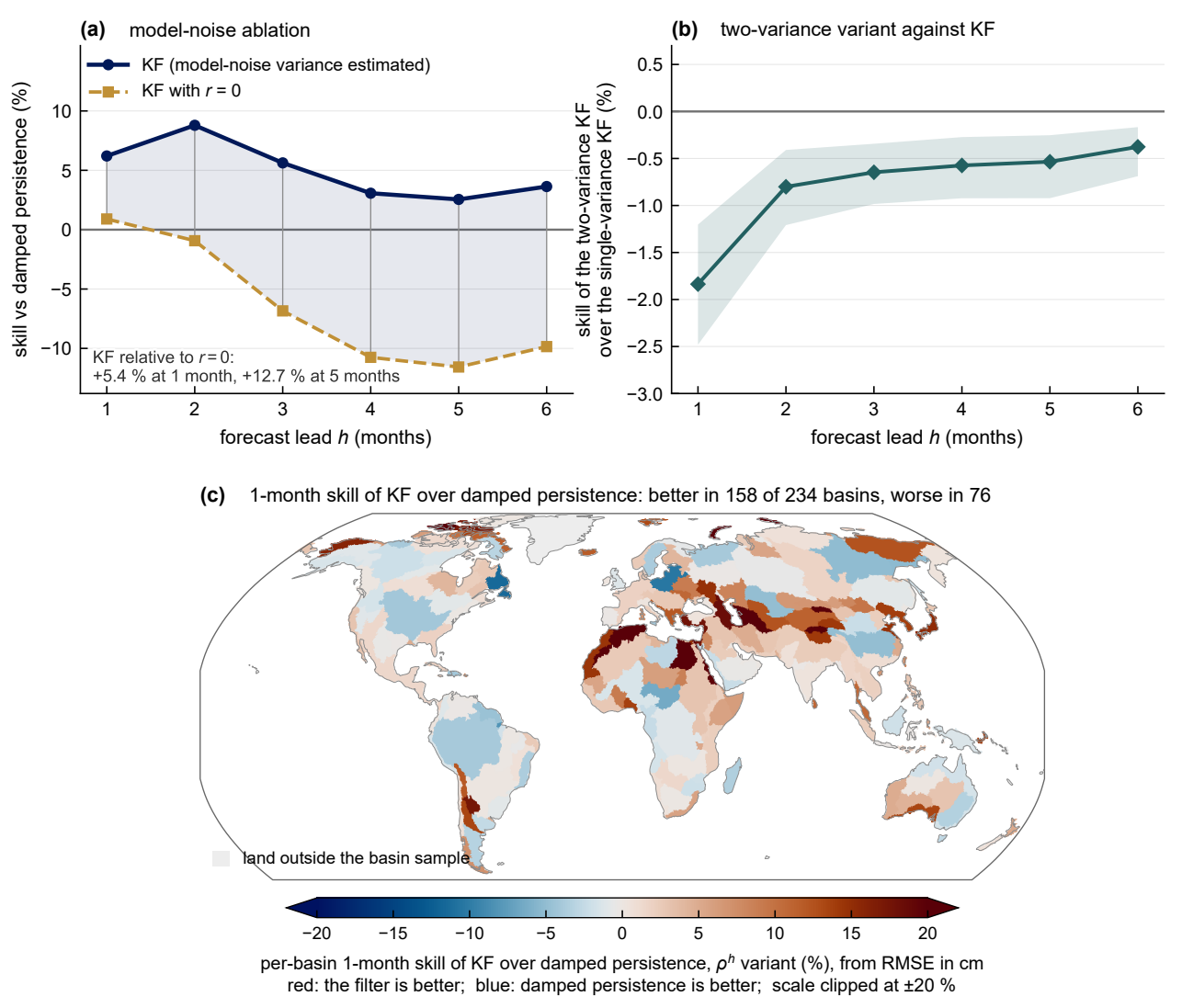

(a) model-noise ablation
KF (model-noise variance estimated)
KF with r = 0
skill vs damped persistence (%)
KF relative to r = 0:
+5.4 % at 1 month, +12.7 % at 5 months
forecast lead h (months)
(b) two-variance variant against KF
skill of the two-variance KF over the single-variance KF (%)
forecast lead h (months)
(c) 1-month skill of KF over damped persistence: better in 158 of 234 basins, worse in 76
land outside the basin sample
per-basin 1-month skill of KF over damped persistence, ρ^h variant (%), from RMSE in cm
red: the filter is better; blue: damped persistence is better; scale clipped at ±20 %